Peter **Olejua** | Biostatistician

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Research Interests

Scalable Bayesian inference (variational Bayes, spike-and-slab variable selection) for high-dimensional and functional regression; functional data analysis of wearable and accelerometer data; longitudinal mixed models for health studies; open-source statistical software in R and C++.

Education

University of South Carolina, Arnold School of Public Health — **Columbia, SC**
Ph.D. in Biostatistics — *Expected May 2027*
Advisor: Dr. Alexander C. McLain. Dissertation on variational Bayes methods for sparse high-dimensional and scalar-on-function regression:

- Parameter expanded variational Bayes for spike-and-slab linear regression (under review).
- Parameter exploded variational Bayes for sparse high-dimensional linear regression (in preparation).
- Variational Bayes for scalar-on-function regression (in progress).

Pontificia Universidad Javeriana — **Bogotá, Colombia**
M.Sc. in Biostatistics — *2020*
Thesis: *TP53 Missense Variants Pathogenicity Prediction Using a Random Forest Meta-Predictor*.

Universidad Simón Bolívar — **Caracas, Venezuela**
B.Sc. in Mathematics — *2013*

Publications

My name is in bold.

Statistical Methodology

[1] A. Zgodic, R. Bai, J. Zhang, **P. Olejua**, and A. C. McLain. "Sparse High-Dimensional Linear Mixed Modeling with a Partitioned Empirical Bayes ECM Algorithm". *Statistics and Computing* 35(4) (2025), 109. DOI: 10.1007/s11222-025-10649-z.

Under Review and Preprints

[1] **P. Olejua**, E. Saha, R. Ghosal, R. Bai, and A. C. McLain. "Parameter Exploded Variational Bayes for Sparse High-Dimensional Linear Regression". 2026. In preparation.

[2] **P. Olejua**, E. Saha, R. Ghosal, R. Bai, and A. C. McLain. *Parameter Expanded Variational Bayes for Well-Calibrated High-Dimensional Linear Regression with Spike-and-Slab Priors*. 2025. DOI: 10.21203/rs.3.rs-7208847/v1. Under review at *Statistics and Computing*.

[3] L. F. Gilón Córdoba, J. F. Betancourt Rodríguez, Á. García, E. A. Cáceres, S. Moreno-Mercado, D. A. Ospina Buitrago, S. Gómez, and **P. Olejua**. *The Accuracy of Two Algorithms of Artificial Intelligence Based on Neural Networks and the CaRDIA-X Algorithm in the Identification of Electronic Implantable Cardiac Devices by Chest X-Rays*. 2022. DOI: 10.22541/au.167117944.47777282/v1.

Applied and Collaborative Research

[1] **P. Olejua**, A. C. McLain, R. Ghosal, E. Regan, and A. Flach. "Longitudinal Study of Changes in Walking and Cognition by Motor Subtype of Parkinson's Disease". *Parkinsonism & Related Disorders* 149 (2026), 108395. DOI: 10.1016/j.parkreldis.2026.108395.

[2] **P. Olejua**, A. C. McLain, N. Inak, M. Dowda, and R. R. Pate. "Clustering Patterns of 24-Hour Physical Activity in Children 6–36 Months Old". *Pediatric Exercise Science* 37(1) (2025), 29–36. DOI: 10.1123/pes.2023-0055.

[3] M. Peñaloza, V. Cuevas, **P. Olejua**, L. Olaya, I. Almonacid, and L. H. Alba. "Factores asociados a la intención de disminuir el consumo de alcohol en personas con consumo riesgoso en un hospital universitario de Bogotá, Colombia". *Revista de la Facultad de Medicina* 71(2) (2023), e98969. DOI: 10.15446/revfacmed.v71n2.98969.

[4] L. H. Alba, M. Peñaloza, **P. Olejua**, E. Cespedes, V. Cuevas, I. Almonacid, L. Olaya, and N. Becerra. "Brief Counseling for Smoking Cessation and Alcohol Use Reduction Concomitant with Hospital Procedures: A Randomized Clinical Trial". *Brazilian Journal of Psychiatry* 44(5) (2022), 507–516. DOI: 10.47626/1516-4446-2021-2413.

[5] S. Barbosa, O. M. Muñoz, A. Cañas Arboleda, Á. A. García, **P. A. Olejua**, and M. Díaz. "Cambio en la mortalidad intrahospitalaria durante el primer pico por SARS-CoV-2 en Bogotá, Colombia". *Infectio* 26(2) (2022), 156–160. DOI: 10.22354/in.v26i2.1015.

[6] M. I. Duran, R. Ballesteros-Ramírez, A. Tellez, L. Torregrosa, **P. A. Olejua**, S. Galvis, C. Urueña, and S. Fiorentino. "Safety Evaluation in Healthy Colombian Volunteers of P2Et Extract Obtained from *Caesalpinia spinosa*: Design 3+3 Phase I Clinical Trial". *Evidence-Based Complementary and Alternative Medicine* 2022 (2022), 7943001. DOI: 10.1155/2022/7943001.

[7] Á. A. García-Peña, E. De-Vries, J. Aldana-Bitar, E. Cáceres, J. Botero, J. Vásquez-Jiménez, R. Tamara, and **P. Olejua**. "Cardiovascular Risk Prediction Models in People Living with HIV in Colombia". *Revista de Investigación Clínica* 74(1) (2022), 23–30. DOI: 10.24875/ric.21000251.

[8] D. Quijano, A. Ali, Y. Arevalo, **P. Olejua**, and J. Trujillo. "Validation of the Spanish Language Version of the Control of Allergic Rhinitis and Asthma Test". *npj Primary Care Respiratory Medicine* 32(1) (2022), 47. DOI: 10.1038/s41533-022-00313-8.

[9] M. Caicedo-Martínez, G. Fernández-Deaza, C. Ordóñez-Reyes, **P. Olejua**, B. Nuche-Berenguer, M. B. Mello, and R. Murillo. "High-Risk Human Papillomavirus Infection among Women Living with HIV in Latin America and the Caribbean: A Systematic Review and Meta-Analysis". *International Journal of STD & AIDS* 32(14) (2021), 1278–1289. DOI: 10.1177/09564624211037498.

[10] J. Chavarriaga, J. Prada, **P. Olejua**, H. López-Ramos, M. Manjarrez, and J.-M. Silva. "Complete Study for Erectile Dysfunction (CompED) Improving Diagnosis and Treatment Decision-Making". *Andrologia* 53(11) (2021), e14212. DOI: 10.1111/and.14212.

[11] C. A. Jiménez Cruz, **P. Olejua**, L. Aguiar Martínez, Á. A. García Peña, J. E. Sotelo Narváez, C. A. Cano Gutiérrez, A. Moreno Carrillo, N. Jurado, A. Garzón, Á. Bustamante, D. Torres, G. París, and M. Santos. "Propuesta de modelo de predicción de mortalidad complementario al triaje en un hospital universitario en Bogotá (Colombia)". *Universitas Médica* 62(3) (2021). DOI: 10.11144/javeriana.umed62-3.tria.

[12] R. Murillo, O. Gamboa, G. Hernández, M. González, **P. Olejua**, M. Molano, and C. Wiesner. "Accuracy of Combined Molecular and Morphology-Based Triage for HPV-Positive Women in Routine Cervical Cancer Screening Services from Colombia". *Preventive Medicine* 153 (2021), 106801. DOI: 10.1016/j.ypmed.2021.106801.

[13] L. Olaya, N. Becerra, V. Cuevas, I. Almonacid, **P. Olejua**, and L. H. Alba. "Características sociodemográficas, familiares y clínicas asociadas con la intención de cesación del tabaquismo en fumadores atendidos en un hospital de cuarto nivel". *Revista de la Facultad de Medicina* 70(3) (2021), e91846. DOI: 10.15446/revfacmed.v70n3.91846.

[14] R. A. Sánchez Forero, **P. A. Olejua**, A. Rocha Morales, and R. Murillo. "Evaluación de errores de posicionamiento en los 6 grados de libertad en pacientes con cáncer de próstata tratados con radioterapia". *Revista Urología Colombiana* 30(1) (2021), 23–33. DOI: 10.1055/s-0040-1714726.

[15] C. F. Valencia, O. D. Lucero, O. C. Castro, A. A. Sanko, and **P. A. Olejua**. "Comparison of ROX and HACOR Scales to Predict High-Flow Nasal Cannula Failure in Patients with SARS-CoV-2 Pneumonia". *Scientific Reports* 11 (2021), 22559. DOI: 10.1038/s41598-021-02078-5.

[16] D. C. Ramírez-Mesías, J. F. Betancourt Rodríguez, Á. A. García, M. L. Juan Guardela, and **P. A. Olejua**. "Características electrocardiográficas del bloqueo de rama izquierda y su capacidad para predecir fracción de eyección disminuida del ventrículo izquierdo". *Revista de la Federación Argentina de Cardiología* (2020).

Software

[1] **P. Olejua** and A. McLain. *spexvb: Parameter Expanded Variational Bayes for High-Dimensional Linear Regression*. Version 0.1.0. CRAN, 2026. DOI: 10.32614/CRAN.package.spexvb.

[2] A. Zgodic, R. Bai, J. Zhang, A. McLain, and **P. Olejua**. *lmmprobe: Sparse High-Dimensional Linear Mixed Modeling with a Partitioned Empirical Bayes ECM Algorithm*. Version 0.1.0. CRAN, 2026. DOI: 10.32614/CRAN.package.lmmprobe.

`spxlvb`: parameter exploded variational Bayes, R package (GitHub). `py-spexvb`: Python port of `spexvb` (GitHub).
GGIR (accelerometer processing, R): two merged pull requests fixing non-wear imputation and activity-log dates (#1522, #1523, 2026).

Research Experience

Department of Epidemiology and Biostatistics, University of South Carolina
Graduate Research Assistant *2021–present*
Supervisor: Dr. Alexander C. McLain.

- Developed parameter-expanded and parameter-exploded variational Bayes algorithms for sparse high-dimensional linear and logistic regression; evaluated them with factorial simulation studies on a Slurm high-performance computing cluster and with genomic data.
- Wrote and maintain the R package `spexvb` on CRAN (Rcpp/RcppArmadillo). Rewrote core routines of `lmmprobe` in C++ with OpenMP.
- Processed wrist accelerometer data (Axivity AX3) with GGIR for pediatric sleep and physical activity studies. Measured the bias of non-wear imputation by imposing real non-wear patterns on complete recordings.
- Fit longitudinal mixed models of gait and cognition in Parkinson's disease and clustering models of 24-hour activity in infants (two first-author papers).

University of South Carolina
Graduate Teaching Assistant *Spring 2023*
BIOS 755, Introduction to Longitudinal and Multilevel Modeling (instructor: Dr. Alexander C. McLain).

Professional Experience

Hospital Universitario San Ignacio **Bogotá, Colombia**
Biostatistician, Research Office *2018–2021*

- Analyzed clinical and observational studies for hospital research groups, leading to co-authored papers in cardiology, infectious disease, psychiatry, and pulmonology.
- Built clinical prediction models for emergency room mortality and for high-flow nasal cannula failure in COVID-19 pneumonia; analyzed adverse events in clinical trials.
- Designed institutional patient registries in REDCap, including the COVID-19 inter-institutional registry.
- Provided biostatistical consulting to internal medicine residents.

Banking and fiduciary sector **Venezuela and Colombia**
Risk Analyst and Consultant *2013–2020*

- Built application and behavioral credit scoring models in SAS at Mercantil Banco Universal (2014–2015) and Banco de Venezuela (2013–2014).
- Segmented money laundering and terrorism financing risk factors with clustering methods for fiduciaries: independent consultant (2017–2020), Fiduciaria Central (2015–2016), Atalaya Consultores (2015).

Professional Service

Referee: *Statistics in Medicine* (2026)

Computing and Methods

Methods: Bayesian inference, variational Bayes, variable selection, functional data analysis, mixed models, survival analysis, clustering, prediction models, simulation studies

Programming: R (package development, Rcpp/RcppArmadillo, testthat), C++, Python, SAS, Stata

Tools: Git/GitHub, LaTeX, Quarto, Slurm, REDCap, GGIR

AI tools: Claude Code, OpenAI Codex (agentic coding for R/C++ package development and analysis pipelines)

Languages: Spanish (native), English (fluent)